

Permit Constraints and Housing Prices: Evidence from U.S. County-Level Data (2021-2022)

Department of Economics

Benjamin Bartel

University of California, Davis

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Abstract.

This paper explores the relationship between permit strictness (measured by a ratio between total residential permits issued and total housing stock) across 344 U.S. county-level jurisdictions from 2021-2022. It finds that a higher permit-to-unit ratio (i.e., more permits issued per unit of housing stock) has a positive but diminishing (convex) effect on housing prices. To isolate this effect, the paper will define six regression models: a simple regression, three multiple linear regressions (MLRs), a Frisch-Waugh-Lovell (FWL) regression, and a one-way fixed-effect regression. This convex behavior suggests that loosening permitting at low levels incentivizes construction or flags counties with robust GDP growth, with diminishing price increases at high ratio levels due to extensive supply. Furthermore, the paper specifies possible endogeneity issues between the permit-to-unit ratio and other variables, especially GDP growth. By extension, it also discusses further research avenues to better identify the true relationship between real estate policy and housing prices.

1. Introduction

From 2000-2024, Americans experienced significant increases in the cost of housing, outpacing real wages, all while the expansion of new housing struggled to keep up with demand (U.S. Department of the Treasury, 2024). This paper investigates the relationship between these high costs and supply constraints. One sensible explanation for the increases relates to urbanization. Cities often provide unique economic opportunities, which are reflected in premium housing costs. Similarly, coastal regions might also demand such a premium, and counties with a higher median income level might have the purchasing power to justify high prices. However, none of these factors fully explain why housing costs in certain regions of the U.S. experience unusually high prices for their specific economic conditions, geography, and urbanization. Because it is the efficient supply and demand model that dictates pricing in real estate markets, real estate asset prices may also reflect supply constraints.

This paper will examine the possible regulatory bottleneck behind these constraints. Specifically, stricter permitting behavior could impair the pace of and incentives for construction, resulting in stagnant or decreasing supply while forcing housing demand to clear the market at higher prices. This paper defines the permit-to-unit ratio as a proxy to capture the extent of the regulatory burden in a given jurisdiction. A high ratio indicates that many permits are issued for each unit of existing housing stock. The reverse is true for a low ratio. In other words, the higher the ratio, the softer the regulatory touch. In short, the paper asks what the effect of the permit-to-unit ratio is on housing prices.

Critically, there are other alternatives to the permit-to-unit ratio that could have served as a proxy for regulatory burden. Indeed, alternative metrics like the Wharton Residential Land Use Regulatory Index (WRLURI) could have been used. However, the permit ratio was underexplored in the literature and is a highly specific metric that theoretically could have an outsized impact on housing prices. See the conclusion for more commentary on opportunities for further research using alternative datasets.

There is ample existing literature related to this study. For instance, Glaeser et al. (2002) identified the adverse impacts of zoning on housing affordability. More recently, Gyourko et al. (2008) introduced the WRLURI to quantify residential land-use regulatory strictness, identifying coastal and urban centers as maintaining the strictest regulatory regimes. Saiz (2010) also identified coastal cities as particularly strict locales, with geography also partially responsible for supply constraints. Paciorek (2013) used the WRLURI to find that builders in heavily regulated cities are less responsive to burgeoning housing demand. Jackson (2016) employed a two-way fixed effect model to examine Californian cities, finding that each incremental land use regulation reduces permits issued by 4%, a supply constraint. Finally, Hsieh and colleagues (2019) identified that American cities, including New York and San Francisco, experience macroeconomic costs due artificially constrained housing supply. Critically, there are few major papers that explore the dynamics of permit policy on housing prices, a gap that this paper seeks to fill.

Other than investigating the causal relationship between permit intensity (as measured by the permit-to-unit ratio) and higher housing prices, the paper will also expand on the above existing literature in describing the endogenous relationship between permitting and growth. As permit policy becomes more lenient in each county, it often accompanies high growth conditions. As will be explored later, the true effect of permitting on prices is not straightforward and direct. Two hypothesized explanations for these counterintuitive results become apparent: a constraint explanation, and a growth explanation. Adding to this texture, there are supply-side and demand-side factors that influence housing prices, tying back to incentives. On the supply side, incentives for builders to increase construction activity could result from looser permitting, high growth, or both. On the demand side, residents could choose a county based on growth conditions, inexpensive prices, or both. In short, permitting could have a murky, if significant, effect on housing prices.

Clarifying the nature of these hypothetical causal relationships has critical policy implications. If the permit-to-unit ratio is an accurate indicator for strong growth conditions in a given county, it could provide high-growth counties with the means to justify housing prices and stay competitive with other countries without these same burgeoning residential costs. Of equal importance is an alternative scenario where high prices are driven by political supply constraints, an opportunity for restrictive countries aiming to unlock growth and improve the cost of living for their residents.

The remainder of the paper will provide more information about the dataset and empirical approach, describe six empirical models, list detailed results, discuss results, summarize the significance of the results, and supply additional material through the appendix and references.

2. Data & Methodology

The Zillow Home Value Index (ZHVI) capturing median home prices was calculated as 12-month averages for 2021 and 2022 (Zillow Group, n.d.). HUD provided permit data for both years (U.S. Department of Housing and Urban Development, n.d.). The U.S. Census computed housing stock July 2020-2024; the 2020-2022 average was taken for 2021 and the 2021-2023 average was taken for 2022) (U.S. Census Bureau, n.d.). It also took median income estimates for both years (U.S. Census Bureau, 2022; U.S. Census Bureau, 2023). The BEA supplied real GDP figures by county for 2021 and 2022, transformed into growth rates (U.S. Bureau of Economic Analysis, n.d.). The USDA produced rural-urban continuum code (RUCC) values by county each decade ranging from 1-9; lower values were more urban. Values were binarized, with 1-3 taking an urban value of 1 and 4-9 taking a rural value of 0. 2023 RUCC values were applied to both years (U.S. Department of Agriculture, Economic Research Service, n.d.). Finally, the U.S. Census produced U.S. coastal counties in 2016 based on 2010 census data (U.S. Census Bureau, 2018). All other counties were assumed to be non-coastal. Coastal counties took value 1 and non-coastal counties took value 0 (National Oceanic and Atmospheric Administration, n.d.). Both years took 2016 data. See Appendix A.2 for more information.

As per Table 5 in A.1, the sample contains 688 observations (344 counties in 2021 and 2022, respectively). The mean of ZHVI was \$315,750.70. The average RATIO value was 1.2%. The mean value for MHI was \$69,723.56. The mean real GDP growth rate was 3.7%. 82.6% of counties were urban, and 14.2% were coastal. Note that these summary statistics were computed jointly using data from both years.

3. Model & Regression Outputs

This paper uses six models to characterize the relationship between permit strictness and housing prices. Full results from all models are recorded in Table 1 of Appendix A.1. All coefficients will be evaluated at the 5% significance level. Also note that all cross-sectional models were evaluated using 2022 data only.

The first model regresses median housing prices (ZHVI) on the permit-to-unit ratio (RATIO). In estimating the value of β_1 , we can use the regression to construct $\hat{\beta}_1$. Per Table 1, the value for the estimate $\hat{\beta}_1$ is 506,000. This suggests that a one-unit increase in RATIO yields an increase in housing prices of \$5,060,000. Below is a formal definition of Model 1. Similar outputs will be provided after each of the other models.

$$ZHVI_i = \beta_0 + \beta_1 \cdot RATIO_i + u_i$$

As this coefficient had a high magnitude, a more robust regression was constructed to minimize omitted variable bias. It introduces four new variables: GROWTH (to measure the real GDP growth rate), MHI (to measure median household income), the quadratic MHI term mhi_sq (to

capture any change in marginal effect on ZHVI as income level changes), and COASTAL (to track coastal status). The parameter of interest remains RATIO's coefficient, estimated by $\hat{\beta}_1$ of 1,960,000 which implies a predicted \$1,960,000 increase in ZHVI per unit increase in RATIO. The regression estimates $\hat{\beta}_2$, $\hat{\beta}_3$, $\hat{\beta}_4$, and $\hat{\beta}_5$ were -257,000 (in dollars per percentage point increase in real GDP growth, not significant), 6.258 (in dollars per dollar in median income), 0 (the square of MHI is not significant, statistically or by magnitude), and 126,000 (in dollars given a county is coastal). Notice that the quadratic MHI term contributed little to the causal inference power of the model.

$$ZHVI_i = \beta_0 + \beta_1 \cdot RATIO_i + \beta_2 \cdot GROWTH_i + \beta_3 \cdot MHI_i + \beta_4 \cdot mhi_sq_i + \beta_5 \cdot COASTAL_i + u_i$$

Model three takes a similar approach to model two and was constructed for the same reasons.

However, it replaces mhi_sq with the quadratic term growth_sq. $\hat{\beta}_1$, $\hat{\beta}_2$, $\hat{\beta}_3$, $\hat{\beta}_4$, and $\hat{\beta}_5$ provide estimates for the coefficients in the formal model depicted below. $\hat{\beta}_1$ remains the parameter of interest. The estimates from the regression are as follows: 2,210,000 (in dollars per incremental permit-to-unit point), -209,000 (in dollars per percentage point increase in real GDP growth, also not significant), -3,430,000 (in dollars per squared unit of GROWTH, also not significant), 6.625 (in dollars per dollar in median income), and 127,000 (in dollars given a county is coastal).

$$ZHVI_i = \beta_0 + \beta_1 \cdot RATIO_i + \beta_2 \cdot GROWTH_i + \beta_3 \cdot growth_sq_i + \beta_4 \cdot MHI_i + \beta_5 \cdot COASTAL_i + u_i$$

The fourth model replaces RATIO with its log-transformed variable ratio_pct and contains the quadratic term ratio_sq_2 while excluding GROWTH and other quadratics for simplicity.

Outputs refer to Table 3. It is the final quadratic model, and the most powerful of the six models.

Like the previous quadratic models, it uses $\hat{\beta}_1$, $\hat{\beta}_2$, $\hat{\beta}_3$, and $\hat{\beta}_4$ to estimate the true coefficients in the model. $\hat{\beta}_1$ remains the parameter of interest. The estimates from the regression are as follows: 32.7% (percentage increase in home prices given a 1 percentage point increase in ratio_pct), -5.4% (similar interpretation, but for ratio_pct's quadratic), 0% (similar interpretation for MHI, only using absolute instead of percentage point terms), and 27.5% (similar interpretation for coastal, given a county is coastal). Critically, all four coefficients are statistically significant at the 5% significance level. This model embodies the convex relationship between RATIO and ZHVI, represented by their log transformations.

$$\ln_zhvi_i = \beta_0 + \beta_1 \cdot \text{ratio_pct}_i + \beta_2 \cdot \text{ratio_sq_2}_i + \beta_3 \cdot \text{MHI}_i + \beta_4 \cdot \text{COASTAL}_i + u_i$$

The fifth model is a composite of two residualized MLR submodels drawing from the logic of the Frisch-Waugh-Lovell theorem. Table 2 contains the output from extracting the residuals. The first residualizes $\ln_zhvi$ via a regression on controls. GROWTH was not statistically significant in the first submodel and was removed.

$$\ln_zhvi_i = \beta_0 + \beta_1 \cdot \text{MHI}_i + \beta_2 \cdot \text{COASTAL}_i + u_i$$

The second submodel residualizes RATIO through a second regression on controls.

$$\text{RATIO}_i = \beta_0 + \beta_1 \cdot \text{MHI}_i + \beta_2 \cdot \text{URBAN}_i + \beta_3 \cdot \text{GROWTH}_i + u_i$$

The two submodels remove any predictable variation from MHI, COASTAL, and (in the case of the second submodel) GROWTH. Both submodels only use 2022 data. After extraction, $\hat{\beta}_1$ was estimated to be 1.136 (measured as percentage point increase in housing prices given an

incremental increase in the permit-to-unit ratio). This estimated effect is not statistically significant. The unified FWL model is depicted below.

$$\text{include_resid}_i = \alpha + \hat{\beta}_1 \cdot \text{PermitRatio_resid}_i + u_i$$

The final model is the FE. It Incorporates ratio_pct, ratio_sq_2, GROWTH, MHI, and a time fixed effect including coastal and urban dummies. Second only to Model Four, it is the most credible model by virtue of its architecture as a fixed effect regression. Its advantage over the cross-sectional models is that it takes time into account. Results are recorded in Table 4. Like all other models, $\hat{\beta}_1$ remains the parameter of interest. Finally, it also uses $\hat{\beta}_2$, $\hat{\beta}_3$, and $\hat{\beta}_4$ to estimate the true coefficients in the model. It also contains δ_t to capture the model's time fixed-effect. Estimates for the coefficients are as follows: 5%, -1%, -11% (not significant), 0 (absolute value, not significant), and 10% (the time fixed-effect). This model also supports the convex relationship found in Model Four.

$$\ln_zhvi_{it} = \alpha_i + \beta_1 \cdot \text{ratio_pct}_{it} + \beta_2 \cdot \text{ratio_sq_2}_{it} + \beta_3 \cdot \text{GROWTH}_{it} + \beta_4 \cdot \text{MHI}_{it} + \delta_t + u_{it}$$

4. Hypothesis Testing Results

This section of the paper provides comprehensive results related to hypothesis testing for outputs from the models defined previously. Generally, visual outputs from models support the conclusion that the permit-to-unit ratio has a convex (positive and diminishing) causal effect on median home prices, as depicted in figures 1 and 3 in Appendix A.1. More commentary on these visual outputs can be found there.

In executing the regressions for each of the models, `RATIO` and `ratio_sq_2` were generally significant. Almost universally across the six models, `RATIO` took a positive value while `ratio_sq_2` took a negative value. This aligns with visual outputs. Many of the controls were generally significant, and Appendix A.1 provides more information.

Because Model Four resulted in the clearest results and causal inference, it was selected for hypothesis testing. Model Six was also selected for hypothesis testing due to being the only fixed-effect model in this study. To determine the significance of the coefficient for `RATIO` in both models, F-tests were conducted for Model Four while Wald tests were conducted for model Six (different due to the regression method used in Model Six). Two single-restriction tests and two multiple-restriction tests were conducted, one of each type for each model (totaling four tests).

The single-restriction F-test for Model 4 tested the null hypothesis that the coefficient of `ratio_pct` was equal to zero. The test gave $F(1, 334) = 69.41 > 3.89$, so the model provides sufficient evidence to reject the null hypothesis at the 5% significance level. Provided below is a formal definition of the null and alternative hypotheses. Similar definitions will follow the remaining three hypothesis tests.

$$H_0^{\text{OLS}} : \beta_1 = 0 \quad \text{vs} \quad H_1^{\text{OLS}} : \beta_1 \neq 0$$

Because Model Six is a fixed-effect regression, it required a Wald test instead of an F test to conduct hypothesis testing. It tested whether the coefficient for `ratio_pct` in Model Six was equal to zero. That said, the single-restriction test for Model Six is displayed in F-format. Per the

regression output in Appendix A.1, the test gave $F(1, 324) = 6.35 > 3.88$, so the model provides sufficient evidence to reject the null hypothesis at the 5% significance level. The formal definition of the null and alternative hypotheses is depicted below.

$$H_0^{\text{FE}} : \beta_1 = 0 \quad \text{vs} \quad H_1^{\text{FE}} : \beta_1 \neq 0$$

Following these two single-restriction hypothesis tests are two multi-restriction hypothesis tests, also corresponding to models four and six, respectively. Both models draw from Table 4 in Appendix A.1. Each test tests the significance of the overall model.

Model Four tests all coefficients against the null hypothesis that all are equal to zero. If at least one coefficient is statistically significant and different than zero, the model offers enough evidence to reject the null hypothesis. If no coefficient meets these criteria, then the model provides insufficient evidence to reject the null hypothesis. Per the Model Four output from Table 4, $F(4, 334) = 153.95 > 2.38$, so the model provides sufficient evidence to reject the null hypothesis at the 5% significance level. The full hypotheses are pictured below.

$$H_0^{\text{OLS}} : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \quad \text{vs} \quad H_1^{\text{OLS}} : \exists j \in \{1, 2, 3, 4\} \text{ such that } \beta_j \neq 0$$

Model Six was tested in a similar manner to Model Four, only using a Wald test instead. Per the Model Six output from Table 4, $\chi_4^2 = 57.19 > 9.488$, so the model provides sufficient evidence to reject the null hypothesis at the 5% significance level. Like the other tests, the hypotheses are pictured below.

$$H_0^{\text{FE}} : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \quad \text{vs} \quad H_1^{\text{FE}} : \exists j \in \{1, 2, 3, 4\} \text{ such that } \beta_j \neq 0$$

6. Conclusion

This paper investigated the relationship between permit strictness and housing prices using county level data from 2021 and 2022. To accomplish this, it used six empirical models: a simple regression, three quadratic MLR models, an FWL regression, and an FE model. Although the FWL model suggested a somewhat flat relationship between the two variables, the other models (especially four and six) provided clear evidence for a convex relationship. At low levels, the permit-to-unit ratio appears to lift housing prices, with a diminishing effect as the ratio increases.

These conclusions come with an important caveat: endogeneity concerns persist. The permit-to-unit ratio and growth move closely together, as shown by Figure 4 in Appendix A.1. From the outset, the ratio was constructed to serve as a proxy gauge for regulatory burden in residential real estate and is influenced by many factors. This potential endogeneity could merit a follow-up study using a different set of variables. Specifically, an independent variable that measures regulatory inefficiency (e.g., the WRLURI) could solve the endogeneity issues endemic to the permit-to-unit ratio. Indeed, a low ratio could indicate slow growth, excessive bureaucracy, or both. Conversely, a high ratio could indicate robust growth, efficient bureaucracy, or both. The risk of endogeneity interferes with the ability to fully understand the relationship between the permit-to-unit ratio and housing prices. Besides swapping regulatory proxies, other changes could include a parallel investigation using city-level datasets to provide more granular results. Conducting the fixed-effect regression over a longer time horizon (longer than two years) could also provide more credible explanatory power. Finally, identifying a valid instrument to predict the permit-to-unit ratio could also quell endogeneity concerns and improve

causal inference without altering the variables in use here. Ideally, this instrument would be strictly defined as a policymaking variable, mostly unresponsive to local economic and demographic conditions.

Despite possible endogeneity concerns, this empirical analysis underscored several critical findings. It demonstrated that a high permit-to-unit ratio eases median housing price growth at high levels. Additionally, as displayed by the relationship between the ratio and real GDP growth rates, fast-growing counties are often counties with high permit-to-unit ratios. These outcomes motivate that thoughtful permitting policy is still critical to promoting housing affordability and spreading prosperity for Americans.

Appendix

A.1 Tables and Figures

Below are nine Stata outputs with commentary below.

Table 1. OLS Regression Results – ZHVI and Permitting (2022)

	(1) Simple OLS	(2) Income Quadratic OLS	(3) Growth Quadratic OLS
Permit Ratio	5.06e+06*** (9.08e+05)	1.96e+06** (6.86e+05)	2.21e+06*** (6.98e+05)
GDP Growth Rate		-2.57e+05 (2.02e+05)	-2.09e+05 (2.04e+05)
GDP Growth Rate ²			-3.43e+06 (2.47e+06)
Median Income		6.258*** (2.019)	6.625*** (0.380)
Median Income ²		0.000 (0.000)	
Coastal Dummy		1.26e+05*** (18401.857)	1.27e+05*** (18361.453)
Constant	2.74e+05*** (14081.684)	-1.70e+05** (82060.152)	-1.82e+05*** (27279.210)
Observations	344	344	344

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2. Residualized Regression – Log Housing Price on Permit Ratio (2022)

	(1)
	ZHVI Residual (Cross-Sectional OLS)
Permit Ratio Residual	1.136 (2.794)
_cons	0.120*** (0.033)
Observations	141

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3. Coefficient Estimates – Ratio Quadratic vs. FE Model

	(1) Ratio Quadratic OLS	(2) Fixed Effects Panel (2021–2022)
Permit Ratio (%)	0.327*** (0.043)	0.028* (0.015)
Permit Ratio ² (%)	-0.054*** (0.010)	-0.005* (0.003)
GDP Growth Rate		-0.094 (0.077)
Median Income	0.000** (0.000)	0.000 (0.000)
Coastal Dummy	0.275*** (0.045)	
Year=0 (2021)		0.000 (.)
Year=1 (2022)		0.099*** (0.004)
Constant	11.095*** (0.068)	12.456*** (0.026)
Observations	344	688

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4. F-Tests and Wald Tests: Models Four and Six

	(1)	(2)
	Ratio Quadratic OLS	Fixed Effects Panel (2021–2022)
Permit Ratio (%)	69.41***	6.35**
Joint Significance (%)	153.95***	57.19***

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5. Summary Statistics (2021 and 2022)

Variable Name	Count	Mean	SD	Min	Max
ZHVI	688	315750.7	169614.3	91566.03	1523473
PERMIT	688	1868.376	3785.815	1	42123
HOUSING	688	134113.8	186356.5	22699	1882526
RATIO	688	0.0123145	0.0112941	0.0000396	0.0822064
MHI	688	69723.56	17448.93	29490	167605
GROWTH	688	0.0370955	0.0378482	-0.1819	0.233
URBAN	688	0.8255814	0.3797451	0	1
COASTAL	688	0.1424419	0.349757	0	1

Figure 1. Permitting Intensity and Log Housing Prices (2022)

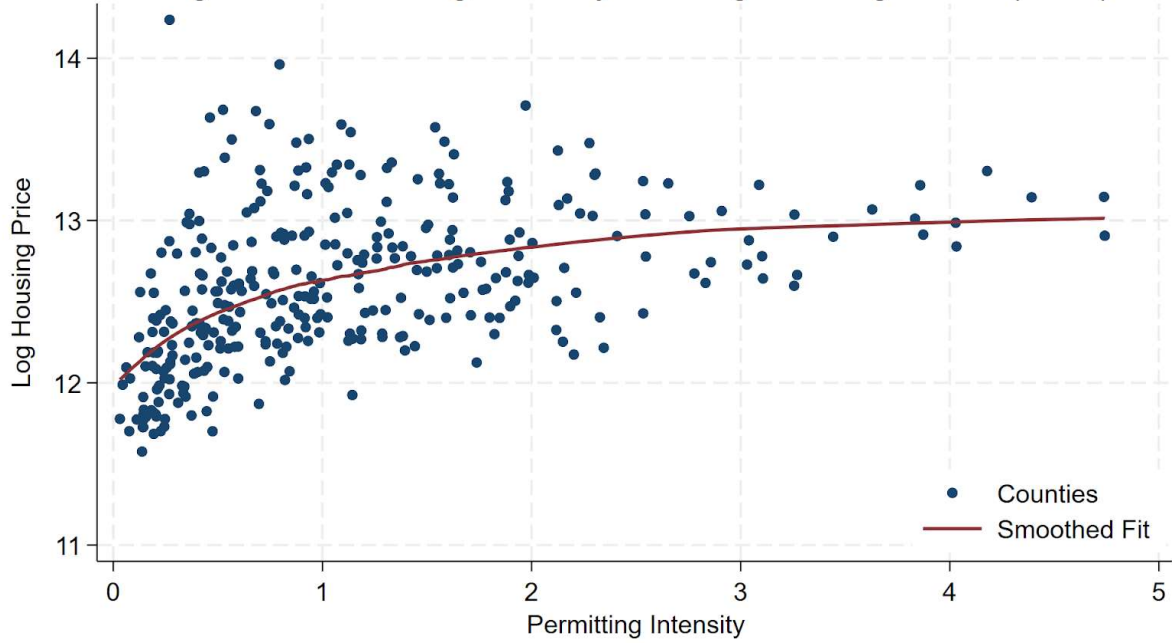


Figure 2. Adjusted Housing Prices vs. Permitting Intensity (2022)

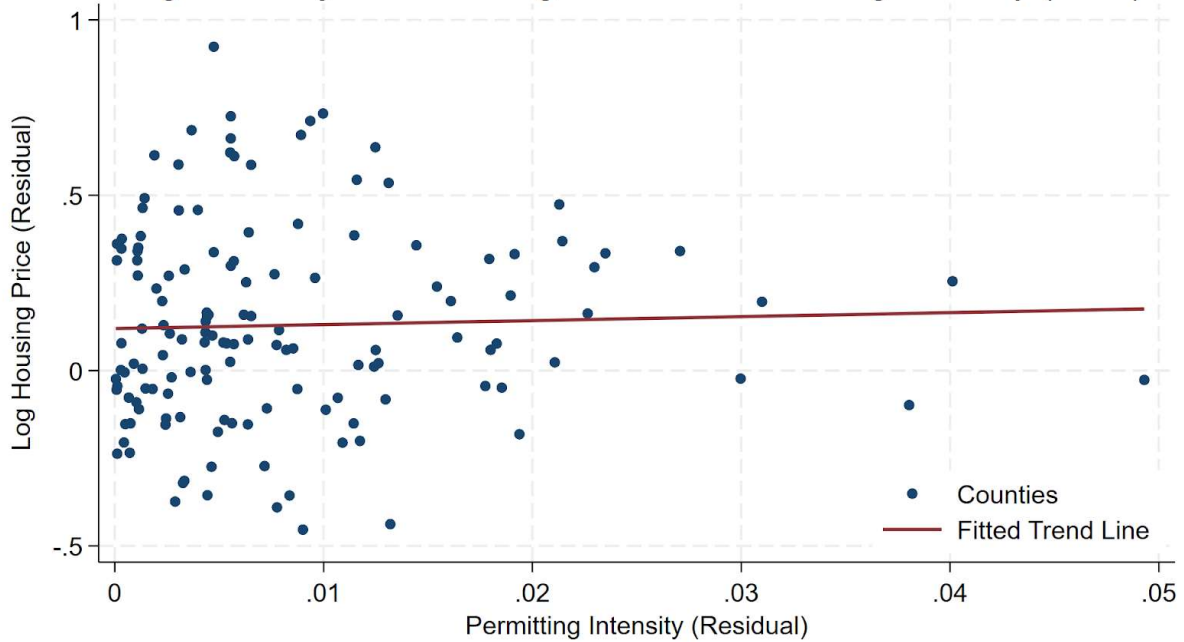


Figure 3. Within-County Residuals vs. Permitting Intensity (2021–2022)

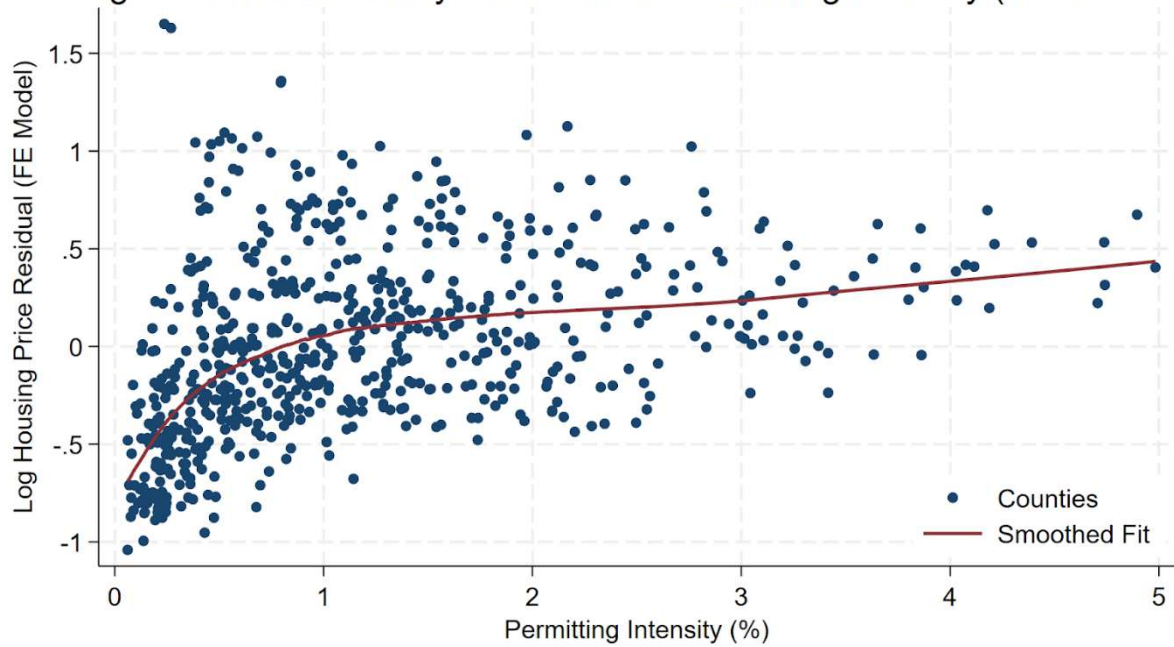
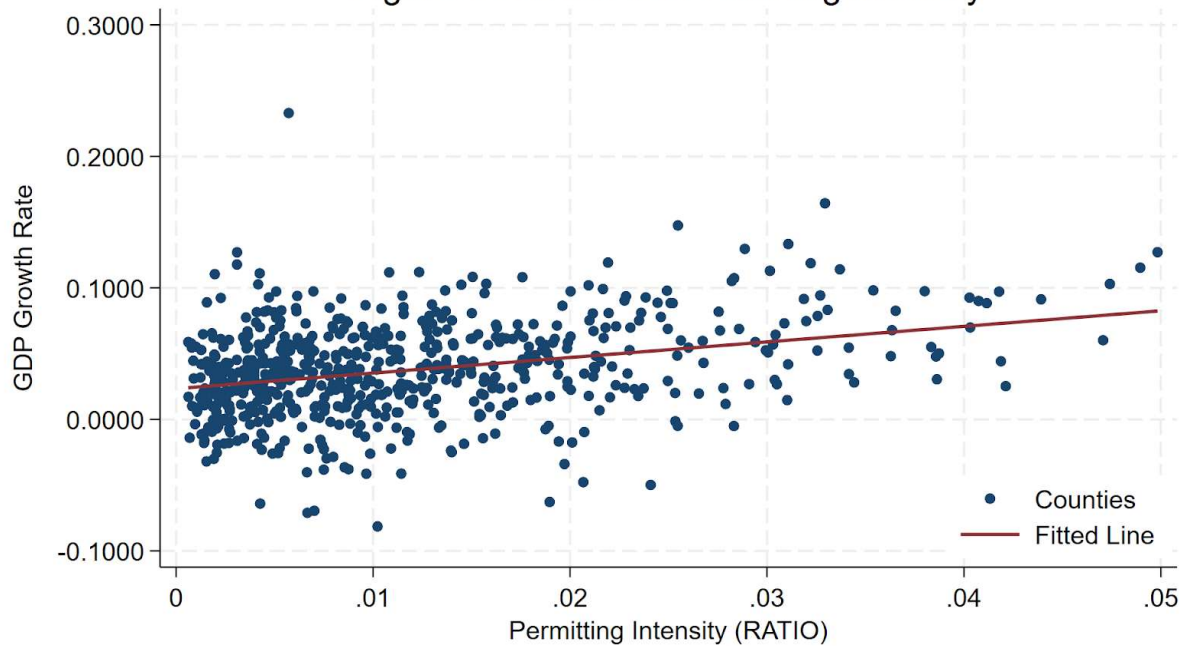


Figure 4. Growth vs. Permitting Intensity



This commentary will interpret each of the nine Stata outputs above. Table 1 includes estimates for constants, coefficients, and standard errors in models one through three. 339 observations were recorded. Growth was generally insignificant, as indicated by a lack of asterisks next to its estimates. Median income was significant across all models, as was the coastal coefficient. The permit ratio was significant across all models. Each model draws from 2022 data.

Table 2 provides the residualized estimated values for the coefficient, standard error, and constant term for model five. Ratio's coefficient is insignificant. The model uses 141 observations from 2022.

Table 3 provides estimates for the coefficients, standard errors, and constant terms for models four and six. The permit-to-unit ratio, recorded as a percentage in both models, was significant. However, the coefficient was only significant at the 10% significance level in Model Six. Ratio's quadratic term was also significant. Model Six, the fixed-effect regression, notably contains many insignificant coefficients. 339 and 676 observations were recorded in the regressions, respectively. Notably, from the quadratic OLS model (Model Four), the coefficient for `ratio_sq_2` = -.05 ($p = .01 < .05$). This is consistent with visual results discussed later, which indicate that the permit-to-unit ratio exhibits a convex effect on housing prices. One possible interpretation for this result is that growth and other factors (mainly on the demand-side) overpower any easing effect that the ratio could have on prices, but that a sufficiently high ratio will allow supply-side factors to depress the positive slope of the relationship between the variables.

Table 4 provides similar outputs to Table 3 but provides F-test outputs. Both models are jointly significant, although the fixed-effect models still struggle to produce significant coefficients. GROWTH and MHI are both insignificant at the 5% significance level. All other coefficients in both models were significant at the 5% level.

Table 5 provides summary statistics for all baseline variables. Summary statistics for transformed variables were excluded to prevent redundancy.

Figure 1 plots Model Four. Permit strictness and prices are drawn directly from a unified regression and demonstrate a clear convex relationship between the two variables. In other words, ratio_pct appears to result in higher housing prices with a diminishing effect as ratio_pct increases.

Figure 2 is associated with Model Two, constructing a scatter plot using the residuals from its two partial regressions. The plot features a weak negative relationship between permit strictness and housing prices.

Figure 3 is like Figure 1. The third scatter plot yields a very similar result to the second. This time, it draws from Model Six, the time fixed-effect regression model. The scatter plot also features a convex relationship between the two variables.

These three models either provide weak evidence or contradict the claim that a higher permit-to-unit ratio decreases housing prices. To investigate why, the fourth scatter plot provides a clear answer: endogeneity concerns. GROWTH and RATIO exhibit a robust positive correlation. In the literature, there is also substantial evidence to defend this relationship (Zhang, Fang, & Xu, 2024). Higher growth rates could lift housing prices, which would in turn cause policymakers to issue more permits per existing housing unit in a bid to ease upward pricing pressures. Put differently, permitting could still ease supply side constraints despite being overpowered by increases in demand.

A.2 More Details on the Dataset

Variable	Description
COUNTY	County
YEAR	Year
ZHVI	Zillow Home Value Index
PERMIT	New Permits Issued
HOUSING	Housing Stock
RATIO	Permit Ratio
MHI	Median Household Income
GROWTH	Real GDP Growth Rate
URBAN	Urban County Indicator
COASTAL	Coastal County Indicator
ln_ZHVI	Log Housing Price
ratio_pct	Ratio (%)
ratio_sq_2	ratio_pct ²
growth_sq	GROWTH ²
mhi_sq	MHI ²
zhvi_resid	ZHVI Residual (Cross-Sectional OLS)
ratio_resid	Permit Ratio Residual (Cross-Sectional OLS)
include_resid	Residual Inclusion Flag (Cross-Sectional OLS)
ln_zhvi_hat	Fitted Log ZHVI (Quadratic OLS)
county_id	County ID (Fixed Effects Panel ID)

ln_ZHVI_hat

Linear Prediction (Fixed Effects Panel)

ln_ZHVI_resid

Residuals (Fixed Effects Panel)

The chart above lists 22 identifiers, variables, and model states. COUNTY is the sole numeric identifier, used to track each of the 344 county observations in the dataset. Beyond identifiers, YEAR tracks whether data is from 2021 or 2022 (taking either value 0 or 1). ZHVI measures the Zillow Home Value Index (median housing price) by county. PERMIT captures the number of new building permits issued, while HOUSING refers to the total housing stock in a given county. RATIO is computed as PERMIT divided by HOUSING. MHI measures median household income. GROWTH reflects real GDP growth rates by county. URBAN takes value 1 if a county is urban and 0 otherwise. Finally, COASTAL takes a value of 1 if a county is located along the coast and 0 otherwise.

Several additional variables were generated in Stata. For interpretability, ratio_pct multiplies RATIO by 100, and ln_ZHVI is the natural log transformation of ZHVI. The variable ratio_sq_2 is the square of ratio_pct, growth_sq is the square of GROWTH, and mhi_sq is the square of MHI. Variables zhvi_resid and ratio_resid store residuals from separate cross-sectional regressions in Model One. Taken from these outputs, include_resid filters for valid, non-outlier residuals to prepare for a plot. Lastly, ln_zhvi_hat contains fitted values from the quadratic OLS model and ln_ZHVI_hat contains fitted values from the fixed effects model. ln_ZHVI_resid stores the residuals from the fixed effects regression for plotting.

In terms of preparation, to provide all six models with a greater ability to isolate the true effect of permit strictness, data was drawn from seven separate sources each representing one variable. The datasets were chosen based on completeness of information and county-level specificity. Selecting these two characteristics maximizes the number of jurisdictions that could be included. City-level data was still widely available but was more challenging to parse through than county-level equivalents. Further investigation using such data could offer more robust results. Data from 2021 and 2022 were selected as they were the most readily available while still being recent enough to provide relevant conclusions for the 2025 housing market in the U.S.

The seven datasets were merged using Google Sheets, matching counties between datasets using Federal Information Processing Standards (FIPS) codes that accompanied each county in the datasets. All counties with missing or incomplete observations from either 2021 or 2022 were omitted from the sample. To simplify the data further, all jurisdictions with repeat names were also omitted. For instance, of the 3,244 county-level jurisdictions in the United States and its territories, “Washington County” appears 30 times (Federal Communications Commission, n.d.).

Once the data was assembled into a single .csv file, it was then imported into Stata for further inspection. Variables and complexity were introduced iteratively to the models to compare regression outputs and find the clearest results.

References

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